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Friction Compensation Using Time Variant Disturbance Observer Based on the LuGre Model

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Abstract— **This paper reports experimental evaluation results of a time variant disturbance observer based on the LuGre friction model for friction compensation. The observer is designed from state space expressions of a controlled plant with the LuGre model that is locally-linearized every sampling period. Thus, observer gains derived from the linearized model become time variant ones as a function of velocity. Parameters of the LuGre model were experimentally identified for a linear stage driven by a ball screw with an AC motor, *i.e.*, a controlled plant. To show the validity of the proposed method, experiments using the controlled plant were conducted to evaluate tracking errors for positioning control with and without the proposed disturbance observer. From the experimental results, it has been confirmed that the proposed disturbance observer can eliminate influence of the friction and is effective to improve positioning accuracy.**

Keywords- LuGre model; Positioning control; Disturbance observer; Friction compensation

I. INTRODUCTION

Nano-scale positioning accuracy has been required for sophisticated mechatronics systems such hard disk drives (HDDs), semiconductor exposure apparatuses and so on. For example, HDDs need more accurate head positioning to achieve higher areal recording density, and semiconductor exposure apparatuses also need higher accuracy to make circuit patterns finer. To attain highly-accurate and high-speed positioning control systems, friction, mechanical resonance, sensing noise could be large problems to be solved. Among those, as Åström pointed it out in [1], friction causes undesired limit cycles and/or offset to the reference and has become one of the largest obstacles against achieving nano-scale positioning. To solve this problem, many methods to compensate/cancel the friction based on various friction models have been proposed [2] - [4].

It is well known that friction has nonlinear dynamics such hysteresis and the Stribeck effect. The Stribeck effect is a dynamic phenomenon that friction force is decreasing as the velocity is increasing for a moment just after an object starts to move. The LuGre model is one of the friction models that can express those nonlinear behaviors [1]. To achieve nano-scale positioning by model-based friction compensation, it is essential to choose as sophisticate models as possible. Thus, the LuGre model gives a good base, and for example, Freidovich *et al*. proposed a friction observer based on the LuGre model, the stability of which is proved via a Lyapunov function [5].

This paper proposes a time variant disturbance observer based on the LuGre model improving the idea in [6]. By analyzing behavior of a linear stage driven by a ball screw and an AC motor, parameters of the LuGre model are identified. After implementing the proposed disturbance observer based on the obtained parameters, experiments of positioning control are conducted to confirm the validity.

II. LUGRE MODEL

In this section, a brief summary of the LuGre model is given, and then a method to determine the LuGre model parameter is explained. The physical meaning of the determined parameter is also discussed associating them with real friction phenomenon.

A. The LuGre model

In this section, characteristics of the LuGre model are described briefly (see [1] for the detail). In the LuGre model, fiction is considered to be generated by deformation of bristles as shown in Fig. 1. Namely, stiffness and viscosity of the bristles, which are modeled as a mass-spring-damper system, generate the friction. Definitions of parameters of the LuGre model and the other parameters of the experimental system are listed in Table I. Friction F_L, which is generated by the model, is given by

$$F_L = \sigma_0 z + \sigma_1 \dot{z} + \sigma_2 v\,, \tag{1}$$

where z is a deformation of the bristles, and its dynamics are described as follows:

$$\dot{z} = v - \sigma_0 \frac{|v|}{\mathrm{g}(v)} z\,, \tag{2}$$

$$\mathrm{g}(v) = F_C + (F_S - F_C) e^{-\left|\frac{v}{v_S}\right|}. \tag{3}$$

In the following part, the term of viscous friction $\sigma_2 v$ is omitted from the friction force to be compensated because it is linear and easy to be compensated. Thus, the friction force here is assumed to be represented as

$$F = \sigma_0 z + \sigma_1 \dot{z}\,, \tag{4}$$

although $\sigma_2 v$ is still incorporated with the state equation below.

B. LuGre model parameter

The experimental system is a linear stage driven by an AC motor via a ball screw (Fig. 2) and it is a typical system that is affected by friction. This section explains how to determine the parameters related to the friction of this system. In this paper, the system is modeled as an equivalent translational one mass

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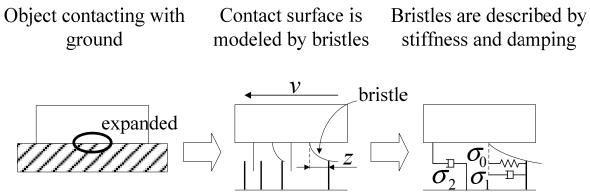

Figure 1. Schematic explanation of the LuGre model.

TABLE I.
DEFINITION OF PARAMETERS AND VARIABLES OF THE LUGRE MODEL.

Definition	Symbol	Value/Unit
Bristle stiffness	σ_0	See Table II.
Bristle damping	σ_1	
Viscous friction coefficient	σ_2	
Coulomb friction force	F_C	
Static friction force	F_S	
Stribeck velocity	v_S	
Torque constant of AC motor	K_a	5.300×10^{-2} N·m/V
Torque/force conversion gain	K_t	8.545×10^{2} N/(N·m)
Mass of system	m	3.270 kg
Position of mass	p	m
Velocity of mass	v	m/s
Bristle deformation	z	m

system as shown in Fig. 3. The motion equation for the equivalent system is given by

$$m\ddot{p} = K_t \cdot K_a \cdot u - F - \sigma_2 v \, .$$

A state space expression of the equivalent system is given by

$$\frac{d}{dt}\begin{bmatrix} p \\ v \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\dfrac{\sigma_1+\sigma_2}{m} & -\dfrac{\sigma_0-\sigma_0\sigma_1\alpha(v)}{m} \\ 0 & 1 & -\sigma_0\alpha(v) \end{bmatrix}\begin{bmatrix} p \\ v \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ \dfrac{K_t\cdot K_a}{m} \\ 0 \end{bmatrix} u , \quad (5)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} p \\ v \\ z \end{bmatrix}, \quad (6)$$

where $\begin{bmatrix} p & v & z \end{bmatrix}^{\mathrm{T}}$ is a state vector, u is the control input in voltage and

$$\alpha(v) = \frac{|v|}{g(v)} .$$

Procedures listed below starts from estimating the friction during the positioning control experiment and ends to determining the LuGre model parameter.

[*Procedure* 1] By using a two degree-of-freedom (DOF) positioning control system with a general learning control, the control input u_{ex} is recorded when the position of the stage almost all perfectly converges to the reference, more specifically when the tracking error becomes less than ±40 pulses in the encoder (0.01/2048 m/pulse). The reference used here is a minimum jerk 5th order function, the distance of which is 2000 pulses (about 4 cm) and the duration time is 0.1 s. The real trajectory of the stage position is also recorded.

[*Procedure* 2] A control input u_{sim} that makes an ideal controlled plant model perfectly track the real trajectory acquired in Procedure 1 is derived by simulation. The ideal controlled plant model here has only the viscous friction $\sigma_2 v$ and no other friction, *i.e.*, F.

[*Procedure* 3] The real friction generated during the positioning control is calculated by subtracting u_{sim} from u_{ex}. The difference of the input u_{ex} and u_{sim} supposes that it is friction which acted on the system at the time of positioning control. Then, parameters of the LuGre model as a function of the stage velocity are determined so as to make the difference between the friction calculated from the model and the real friction, $u_{ex}-u_{sim}$, as small as possible.

The above described procedures are to fit the time response of the friction from the model during relatively fast motion. To confirm that the determined parameters are appropriate in slow motion, the hysteresis characteristics are also checked. Namely, hysteresis loops between in the real system and in the simulation system with the derived friction model are compared during the stage position is tracking a sinusoidal input of 0.2 Hz, the amplitude of which is 100 pulses.

Table II list the determined parameters derived from the above procedures. Figure 4 shows comparison result of time responses of the actual friction, $u_{ex}-u_{sim}$, and the friction calculated from the determined parameter in the moving distance of 2000 pulses. Hysteresis loops for the 0.2 Hz sinusoidal input are also compared with each other (Fig. 5). Note that the unit used in Figs. 4 and 5 is Newton that is converted from voltage, which is the unit of u_{ex} and u_{sim}.

A fitting rate defined as

$$Fit = \left(1 - \frac{\sqrt{\sum_{k=1}^{N}\left(\hat{y}(k)-y(k)\right)^2}}{\sqrt{\sum_{k=1}^{N}\left(y(k)-\bar{y}\right)^2}} \right) \times 100 \, [\%],$$

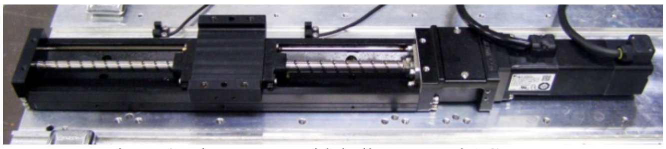

Figure 2. Linear stage with ball screw and AC motor.

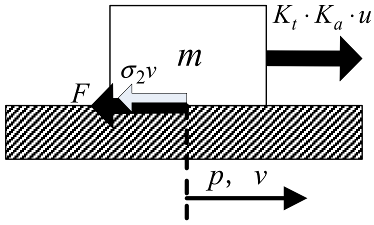

Figure 3. Simplified model of linear stage.

TABLE II. PARAMETER OF THE LUGRE MODEL.

Parameter	value
σ_0	5.000×10^{4} N/m
σ_1	6.000×10^{2} kg/s
σ_2	93.09 kg/s
F_C	3.500 N
F_S	9.000 N
v_S	7.500×10^{-2} m/s

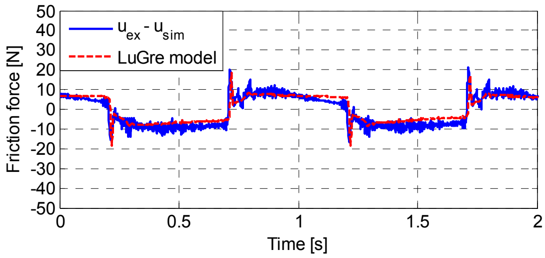

Figure 4. Comparison of real and modeled friction (2000 pulse).

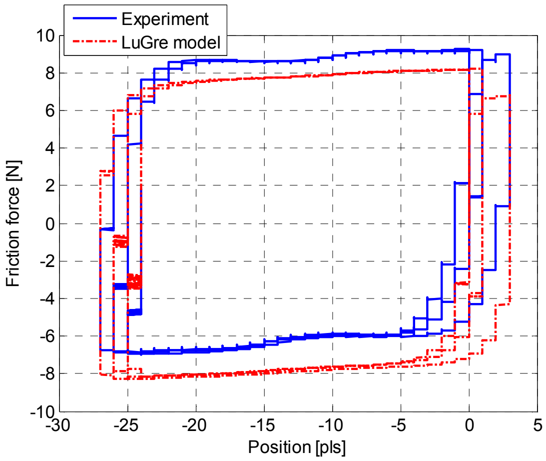

Figure 5. Hysteresis characteristic for friction.

is calculated for the result in Fig. 4, where $y(k)$ is an actual time response (reference), the mean of which is $\bar{y}(k)$, and $\hat{y}(k)$ is the corresponding modeled response. The calculated fitting rate for Fig. 4 was 59.6 %. This rate is not high enough to use the model by itself in generating a feedforward manner.

However, since the determined LuGre model is used as a base model of an observer with an error correction feedback, the characteristics for both the time response (Fig. 4) and the hysteresis loop (Fig. 5) can be said to well represent the actual nonlinear dynamics of friction.

In order to verify the validity of the determined parameters, time responses in different moving distances, *i.e.*, 6000 and 8000 pulses were also examined. Comparisons between time responses of the friction generated by the determined LuGre model and the actual friction are shown in Figs. 6 and 7. The fitting rates were respectively 37.8 % and 19.8 %. Though the fitting rates decrease as the moving distances increase, it has been confirmed by experiments as discussed later in Section IV that the proposed observer base on the determined LuGre model parameters works well for all the moving distances.

III. Time Variant Disturbance Observer Based on the LuGre Model

In this section, the proposed time variant disturbance observer based on the LuGre model is explained after quickly reviewed the continuous time case (see [6] - [8] for details). The disturbance observer is hereinafter called "time variant DOB."

As described in [6], the continuous-time friction observer is given by

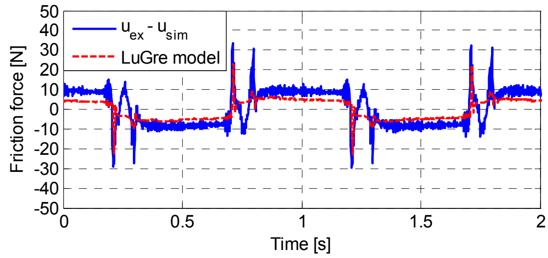

Figure 6. Comparison of real and modeled friction (6000 pulse).

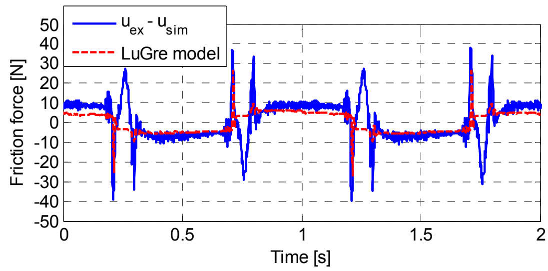

Figure 7. Comparison of real and modeled friction (8000 pulse).

$$\frac{d}{dt}\begin{bmatrix}\hat{p}\\ \hat{v}\\ \hat{z}\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & -\dfrac{\sigma_{1m}+\sigma_{2m}}{m} & -\dfrac{\sigma_{0m}-\sigma_{0m}\sigma_{1m}\alpha(\hat{v})}{m}\\ 0 & 1 & -\sigma_{0m}\alpha(\hat{v})\end{bmatrix}\begin{bmatrix}\hat{p}\\ \hat{v}\\ \hat{z}\end{bmatrix} + \begin{bmatrix}0\\ \dfrac{K_t\cdot K_a}{m}\\ 0\end{bmatrix}u+\begin{bmatrix}k_1(\hat{v})\\ k_2(\hat{v})\\ k_3(\hat{v})\end{bmatrix}(p-\hat{p}) \quad ,(7)$$

$$\hat{y}=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}\hat{p}\\ \hat{v}\\ \hat{z}\end{bmatrix}, \quad (8)$$

where the subscript m shows that the parameters are the identified ones. In designing the time variant DOB, the observer gain $[k_1(\hat{v}) \;\; k_2(\hat{v}) \;\; k_3(\hat{v})]^{\mathrm{T}}$ is derived in the framework of the steady state Kalman filter by solving the state dependent Riccati equation (SDRE) assuming that the nonlinear term $\alpha(\hat{v})$ can be replaced by a constant at the sampling instant, where the process noise covariance Q and the sensor noise covariance R are used as tuning parameters.

Based on the continuous time case, a discrete-time observer is proposed. Since the continuous-time controlled plant, (5) and (6), can be separated into the time invariant subsystem and the time variant one, the controlled plant is separately discretized assuming the use of a zero-order hold. Then, the two discrete-time subsystems are consolidated again to build a whole discrete-time observer. The time invariant subsystem to be used a base model of the observer is given by

$$\begin{aligned}\frac{d}{dt}\hat{\boldsymbol{x}}_m &= \begin{bmatrix}0 & 1\\ 0 & -\dfrac{\sigma_{2m}}{m}\end{bmatrix}\hat{\boldsymbol{x}}_m+\begin{bmatrix}0\\ \dfrac{K_t\cdot K_a}{m}\end{bmatrix}u\\ &= \boldsymbol{A}_m\hat{\boldsymbol{x}}_m+\boldsymbol{B}_m u\end{aligned} \quad (9)$$

where the state vector is $\hat{\boldsymbol{x}}_m=[\hat{p} \;\; \hat{v}]^{\mathrm{T}}$. Equation (9) is easily discretized, and the system, the state vector of which is

$\hat{\boldsymbol{x}}_{md} = [\hat{p}_d(k) \quad \hat{v}_d(k)]^{\mathrm{T}}$, is obtained as

$$\hat{\boldsymbol{x}}_{md}(k+1) = \boldsymbol{A}_{md}\hat{\boldsymbol{x}}_{md}(k) + \boldsymbol{B}_{md}u_d(k), \tag{10}$$

where

$$\boldsymbol{A}_{md} = e^{A_m T_s}, \quad \boldsymbol{B}_{md} = \int_0^{T_s} e^{A_m \tau}\boldsymbol{B}_m d\tau,$$

and T_s is the sampling interval, k is the time index to show the k^{th} sampling interval. The time variant subsystem, the dynamics of which is defined by (2), is discretized as follows:

$$\hat{z}_d(k+1) = e^{-\sigma_{0m}\frac{|\hat{v}_d(k)|}{g(\hat{v}_d(k))}T_s}\hat{z}_d(k) - \frac{g(\hat{v}_d(k))}{\sigma_0|\hat{v}_d(k)|}\left(e^{-\sigma_0\frac{|\hat{v}_d(k)|}{g(\hat{v}_d(k))}T_s} - 1\right)\hat{v}_d(k). \tag{11}$$

Eventually, the consolidated system, the states of which are $[\hat{p}_d(k) \quad \hat{v}_d(k) \quad \hat{z}_d(k)]^{\mathrm{T}}$, is given by

$$\begin{bmatrix}\hat{p}_d(k+1)\\ \hat{v}_d(k+1)\\ \hat{z}_d(k+1)\end{bmatrix} =$$

$$\begin{bmatrix} \boldsymbol{A}_{md} + \begin{bmatrix}0 & -\boldsymbol{B}_{mgd}\sigma_{1m}\\ 0 & \end{bmatrix} & -\boldsymbol{B}_{mgd}(\sigma_{0m} - \sigma_{0m}\sigma_{1m}\alpha(\hat{v}_d(k))) \\ 0 \quad -\frac{g(\hat{v}_d(k))}{\sigma_{0m}|\hat{v}_d(k)|}\left(e^{-\sigma_{0m}\frac{|\hat{v}_d(k)|}{g(\hat{v}_d(k))}T_s} - 1\right) & e^{-\sigma_{0m}\frac{|\hat{v}_d(k)|}{g(\hat{v}_d(k))}T_s} \end{bmatrix}\begin{bmatrix}\hat{p}_d(k)\\ \hat{v}_d(k)\\ \hat{z}_d(k)\end{bmatrix}, \tag{12}$$

$$+\begin{bmatrix}\boldsymbol{B}_{md}\\ 0\end{bmatrix}u_d(k) + \begin{bmatrix}k_{d1}(\hat{v}_d(k))\\ k_{d2}(\hat{v}_d(k))\\ k_{d3}(\hat{v}_d(k))\end{bmatrix}(p_d(k) - \hat{p}_d(k))$$

$$\hat{y}_d(k) = [1 \quad 0 \quad 0]\begin{bmatrix}\hat{p}_d(k)\\ \hat{v}_d(k)\\ \hat{z}_d(k)\end{bmatrix}, \tag{13}$$

where

$$\boldsymbol{B}_{mgd} = \int_0^{T_s} e^{A_m\tau}\begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix}d\tau.$$

Thus, the estimated friction is obtained as

$$\hat{F}_d(k) = [0 \quad \sigma_{1m} \quad \sigma_{0m} - \sigma_{0m}\sigma_{1m}\alpha(\hat{v}_d(k))]\begin{bmatrix}\hat{p}_d(k)\\ \hat{v}_d(k)\\ \hat{z}_d(k)\end{bmatrix}. \tag{14}$$

The observer gain is derived in the same way as the continuous time case although the discrete-time Riccati equation is used. The observability matrix becomes singular in some points depending on the nonlinear term $\alpha(\hat{v}_d(k))$. Since the system is unobservable but is still detectable at the points, around the unobservable area, the observer gain is interpolated by using quadratic function so that all the observer's poles are placed to appropriate area inside the unit disc.

IV. Experiment for Positioning Control

In order to verify the validity of the proposed friction compensator, a 2DOF positioning control system is designed, and by using the 2DOF positioning control system in combination with three different observers that include the proposed one, positioning performances are evaluated.

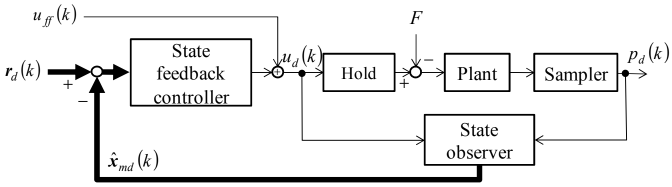

Figure 8. Two DOF positioning control system without friction compensation.

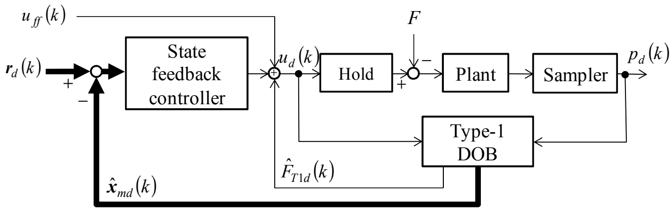

Figure 9. Two DOF positioning control system using type-1 DOB.

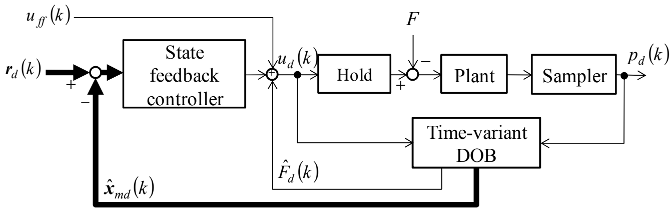

Figure 10. Two DOF positioning control system using time variant DOB based on the LuGre model.

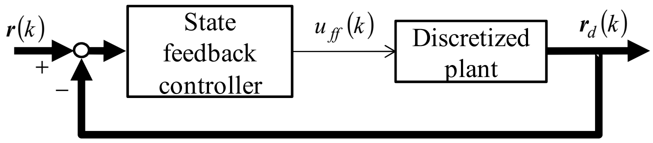

Figure 11. Reference trajectory generator.

A. Positioning control system setup

Three 2DOF positioning control systems are implemented using three different observers. The first one uses a common second order observer to estimate only position and velocity and has no function to compensate friction. The second one has a type-1 DOB to compensate friction. The third one is to use the proposed time variant DOB. Since the type-1 DOB is one of the most commonly used compensator to cancel the friction, it is proper to compare the proposed method with it. Figures 8, 9 and 10 show the block diagram of each 2DOF positioning control system. In the figures, $r_{pd}(k)$ is the reference position trajectory, $r_{vd}(k)$ is the reference velocity trajectory, and $u_{ff}(k)$ is the feedforward control input. Thus, $\boldsymbol{r}_d(k)$ is defined as $\boldsymbol{r}_d(k) = [r_{pd}(k) \quad r_{vd}(k)]^{\mathrm{T}}$.

As shown in Fig. 11, a reference trajectory generator that consist of a discretized controlled plant driven by a state feedback controller to make the reference model follow a minimum jerk trajectory $\boldsymbol{r}(k) = [r_p(k) \quad r_v(k)]^{\mathrm{T}}$ as a 5^{th} order function of time, the positioning time of which is 0.1 s. The position and velocity generated by the reference model are used as $\boldsymbol{r}_d(k)$, and the control input is used as the feedforward input $u_{ff}(k)$ for the 2DOF positioning control systems.

The state feedback gains and the observer gains are obtained by the Linear-Quadratic-Gaussian (LQG) design. As explained above for the time variant DOB case, the observer

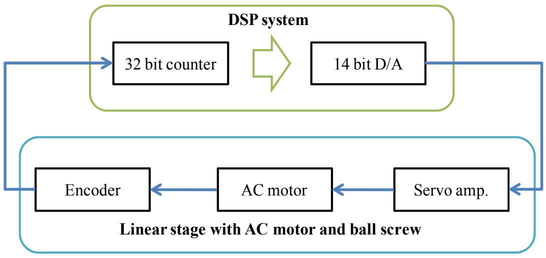

Fig. 12: Experimental setup.

TABLE III. THE WEIGHTING MATRICES.

Controller and Observer	Weighting Q	Weighting R
State feedback controller	diag[2×10^{6} 3]	0.3
State observer	diag[1×10^{-2} 1]	0.3
Type-1 DOB	diag[1×10^{-2} 1 7×10^{4}]	0.3
Time variant DOB	diag[1×10^{-3} 1×10^{-2} 1×10^{2}]	1

TABLE IV. DESIGN PARAMETERS OF OBSERVER AND CONTROLLER.

Description	State observer	Type-1 DOB	Time variant DOB
Gain k_{d1}	1.689×10^{-1}	1.866×10^{-1}	Function of $\hat{v}_d(k)$
Gain k_{d2}	3.685×10^{-1}	3.485	
Gain k_{d3}	—	4.366×10^{2}	
Gain f_{md1}	2.335×10^{3}		
Gain f_{md2}	2.087×10^{1}		

gains of the second order observer and the type-1 DOB are also derived by using the process noise covariance Q and the sensor noise covariance R as tuning parameters.

B. Experimetal conditions

The controlled plant is a linear stage driven by an AC motor via a ball screw (Fig. 2), and Fig. 12 shows the whole experimental system, the sampling time of which is 1 ms.

Base on he identified LuGre model parameters in Table II and the weighting matrices in Table III, controller gains were obtained. For the type-1 DOB, a weight for the friction was tuned to make the tracking error as small as possible in the range where undesirable oscillations did not occur. The resulting observer gains, k_{d1}, k_{d2} and k_{d3}, the state feedback gains, f_{md1} and f_{md2}, are listed in Table IV. The reference trajectories were designed for the moving distances of 2000 pulses (about 1 cm), 6000 pulses (about 3 cm) and 8000 pulses (about 4 cm) with a period of 0.5 s, which include the moving time of 0.1 s and the stationary time of 0.4 s.

C. Experimental result

Figures 13, 14 and 15 show time responses in positioning control of 2000 pulse displacement respectively for the cases of no friction compensation, a type-1 DOB and the proposed time variant DOB. For each figure, two plots from the top shows respectively time responses of reference and actual trajectories and the tracking error. For Figs. 14 and 15, a time response of estimated friction is shown as the third plot. The forth plot in Fig. 15 shows tracking error in an expanded scale. Time responses in Figs. 13, 14 and 15 are typical ones of 20 trials, and all the other responses was almost all the same as those.

As shown in Fig. 13, in the case of no friction, there was about 150 pulse steady-state error remained due to the friction since there was no integrator in the feedback controller.

Figure 14 shows that the steady-state error can be eliminated by using a type-1 DOB although the convergence

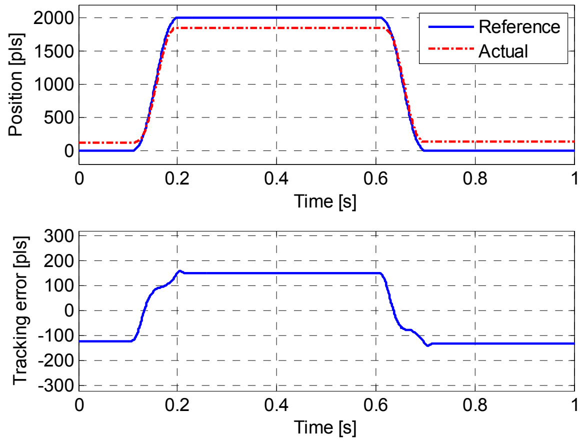

Fig. 13. Time response of two DOF control system without friction compensation: from the top, position and tracking error (2000 pulse).

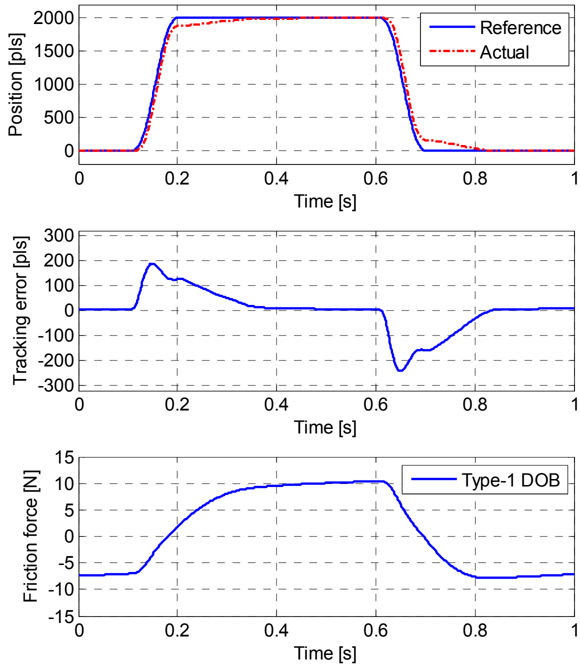

Fig. 14. Time response of two DOF control system with type-1 DOB: from the top, position and tracking error, estimated friction (2000 pulse).

was relatively slow. This is because the type-1 DOB compensation is essentially the same as that of an integrator and is not expected to provide fast friction estimation even after feedback gain tuning. This slow response also caused large peaks in tracking error more than 200 pulses.

As shown in Fig. 15, by using the proposed time variant DOB, the tracking error was suppressed to be less than ±25 pulses with no steady-state error. By comparing the third plots in Figs. 14 and 15, it has been confirmed that the proposed observer can estimate the friction faster than the type-1 DOB. This feature keeps the tracking error during positioning small.

For longer moving distances, *i.e.*, 6000 and 8000 pulses, only the time responses of tracking error are plotted. Figures 16 and 18 show the type-1 DOB case, and Figs. 17 and 19 show the proposed time variant DOB case. The proposed method can reduce the peak value of tracking error by 80 % compared with

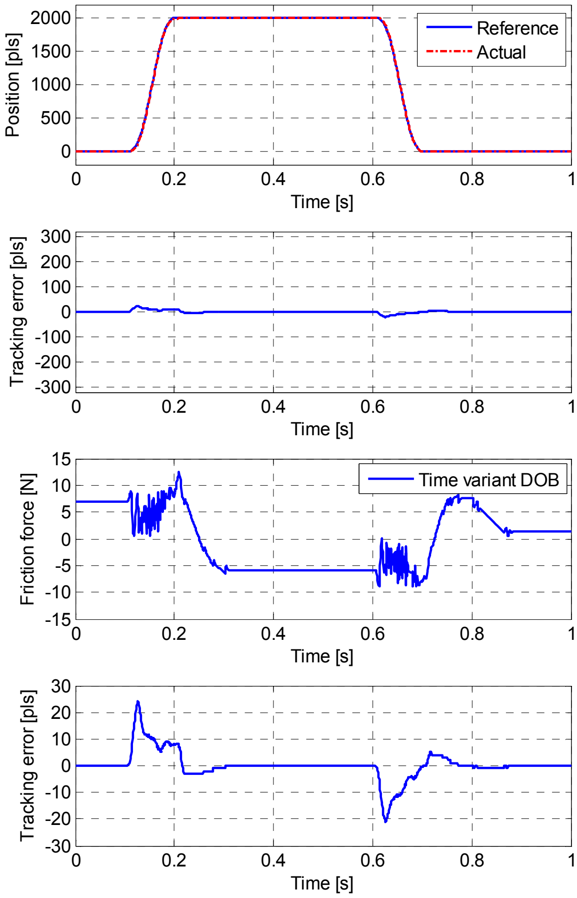

Fig. 15. Time response of two DOF control system with time variant DOB: from the top, position and tracking error, estimated friction, expanded tracking error (2000 pulse).

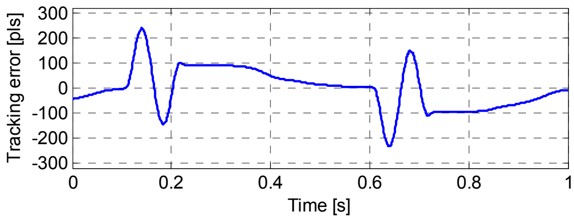

Fig. 16. Time response of two DOF control system with type-1 DOB: tracking error (6000 pulse).

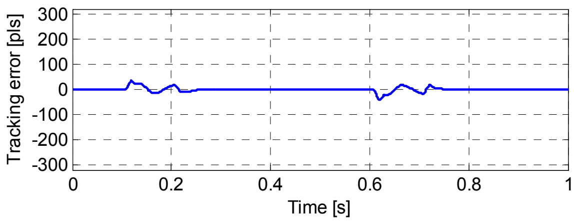

Fig. 17. Time response of two DOF control system with time variant DOB: tracking error (6000 pulse).

that of the type-1 DOB case although the tracking error for both cases increases as the moving distance increases.

Thus, it has confirmed that the time variant DOB is effective to eliminate influence of the friction and to reduce tracking error in positioning control.

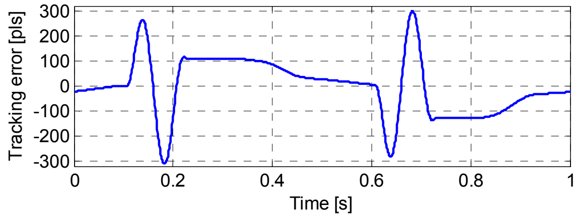

Fig. 18. Time response of two DOF control system with type-1 DOB: tracking error (8000 pulse).

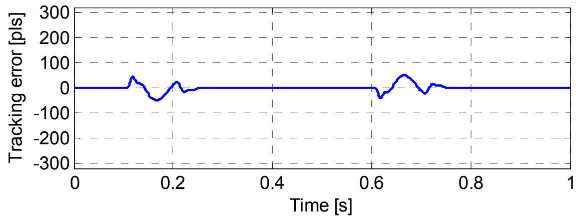

Fig. 19. Time response of two DOF control system with time variant DOB: tracking error (8000 pulse).

V. CONCLUSION

This paper proposed a time variant disturbance observer (DOB) based on the LuGre friction model to compensate the friction and to improve positioning control performance. The parameters of the LuGre model that is experimentally identified was used to design the observer. A discrete-time 2DOF controller with the proposed time variant DOB was implemented to a linear stage driven by a ball screw with an AC motor to evaluate the validity of the proposed system by experiment. Experimental results comparing it with some conventional methods showed that the proposed method is effective to reduce influence of friction.

Future work will be to develop a method to derive optimal parameters of the LuGre model that minimized the fitting rate via nonlinear identification method.

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